📄 Project Concept & Vision

Project Idea

This project is a **local on-demand delivery platform** that connects customers, local stores, and independent drivers in one system.

It is inspired by global solutions like Uber Eats and Instacart, but designed specifically for the needs of our region.

Definition

The platform is a **multi-sided marketplace system** that enables:

- Customers to order products from nearby stores
- Stores to sell and manage orders digitally
- Drivers to earn money by delivering orders

All in real-time, using mobile applications and a central backend system.

Project Goals

1. 💼 Economic Empowerment

- Create **job opportunities** for drivers (especially youth)
- Help people earn flexible income (part-time or full-time)

2. 🏪 Support Local Businesses

- Enable small shops to go digital
- Increase their sales without needing complex technology

3. 🚀 Improve Daily Life

- Make ordering groceries and goods **fast and easy**
- Reduce the need to travel or wait in stores

4. 🌍 Local Innovation

- Build a system tailored to **our region's reality**:
  - Cash payments
  - Map-based locations
  - Motorbike delivery

🤝 What the Platform Provides

👤 For Customers

- Easy ordering from nearby stores
- Live tracking of delivery
- Flexible payment (cash / future digital payments)

🏪 For Store Owners

- Digital store (no technical skills required)
- Order management system
- More customers and visibility

🚴 For Drivers

- Flexible work (choose when to work)
- Earn money per delivery
- No boss — independent work

💡 Value Proposition (Why this project matters)

This platform solves real problems in the local market:

- ❌ Lack of organized delivery systems
- ❌ Limited job opportunities

- ❌ Small shops not digitized

👉 By solving these, the project creates a **win-win ecosystem**:

- Customers save time
- Drivers earn money
- Stores grow their business

💰 Business Goal

The project is also designed to be profitable by:

- Taking a Subscribtions
- Delivery fees for delivery man
- Optional premium services (later)

🚀 Long-Term Vision

- Expand to multiple cities
- Add digital payments & wallets
- Build a strong logistics network
- Become a leading local delivery platform

🧾 Short Version

> A mobile-based delivery platform that connects customers, local stores, and drivers to enable fast, reliable, and affordable delivery services, while creating income opportunities and supporting local businesses.

📄 Software Requirements Specification (SRS)

📄 Use Cases for Your Delivery System

❌The Core Problem : collect Money 💰

for the problem of collect money as bussines owner i will use subscribtion approuch that depends on work of store and delivery man with many parameters with easy subscribtion for those how entry the app.

📦 System Architecture & Technology Stack

Pages:

Admin page

Admin Page

Customer Page

Cutomer Page

Delivery Man Page

Delivery man Page

Store Page

Store Page

🧠 Overview

This project is a **subscription-based local delivery platform** that connects customers, stores, and delivery drivers through a scalable microservices architecture.

🏗️ System Architecture

🌐 API Gateway

- Acts as a single entry point for all client requests
- Handles:
  - Authentication (JWT validation)
  - Routing to microservices
  - Security and rate limiting

⚙️ Microservices

🔐 Auth Service

- User authentication (login/register)
- JWT token generation and validation
- Role management (Customer, Store, Driver, Admin)

👥 User Service

- Manages all users:
  - Customers
  - Drivers
  - Store owners
- Profile management

🏪 Store Service

- Store management
- Product catalog (CRUD)
- Store order handling

📦 Order Service

- Create and manage orders
- Order status lifecycle
- Central business logic for orders

🚴 Delivery Service

- Driver availability
- Order assignment
- Delivery tracking

💳 Subscription Service

- Manage subscriptions for:
  - Stores
  - Drivers
- Subscription plans and expiration

⭐ Rating Service

- Store and driver ratings
- Reviews management
- Visibility scoring logic

📬 Message Broker

- Handles asynchronous communication between services
- Event-driven architecture

Technology Options:

- RabbitMQ (recommended for MVP)
- Apache Kafka (for scaling)

🗄️ Databases

- Each microservice owns its own database

Suggested DB:

- PostgreSQL

Examples:

- auth_db
- user_db
- order_db
- store_db
- subscription_db

📱 Client Applications

👤 Customer App

- Browse stores
- Place orders
- Track deliveries

🚴 Driver App

- Receive delivery requests
- Accept/reject orders
- Navigation

🏪 Store Dashboard

- Manage products
- Accept/reject orders

- Track sales

🛠️ Admin Panel

- Manage users
- Monitor system
- Handle reports and issues

🧰 Technology Stack

💻 Backend

- Java
- Spring Boot
- Spring Cloud (optional for scaling)

📱 Mobile

- Android (Java / Kotlin)

🌐 Frontend (Optional Web)

- React.js or Angular

🗄️ Database

- PostgreSQL

📬 Messaging

- RabbitMQ (start)
- Apache Kafka (future scaling)

🐳 Containerization

- Docker

📦 Orchestration

- Docker Compose (MVP)
- Kubernetes (future)

🌐 API Gateway

- Spring Cloud Gateway or NGINX

🔔 Notifications

- Firebase Cloud Messaging (FCM)

📍 Maps & Location

- Google Maps API

🚀 Deployment Architecture

🖥️ Servers

- VPS (start):
  - DigitalOcean / Hetzner
- Cloud (later):
  - AWS / Google Cloud

🐳 Deployment Strategy

Phase 1 (MVP)

- Single VPS
- Docker Compose
- All services containerized

Phase 2 (Scaling)

- Multiple servers
- Kubernetes cluster
- Load balancing

🌍 Domain & Networking

- Domain name (e.g., yourapp.com)
- Subdomains:
  - api.yourapp.com
  - admin.yourapp.com

🔐 Security

- JWT authentication
- HTTPS (SSL via Let's Encrypt)

📊 Monitoring & Logging

Basic (Start)

- Docker logs

Advanced (Later)

- Prometheus (monitoring)

- Grafana (visualization)

🔁 CI/CD (Future)

- GitHub Actions
- Automated build & deployment pipelines

🧠 Architecture Principles

- Microservices-based architecture
- Event-driven communication
- Loose coupling between services
- Database per service
- API Gateway pattern

🚀 Future Enhancements

- Digital payment integration
- Wallet system
- Advanced analytics
- AI-based driver assignment
- Multi-city scaling

📌 Summary

This system is designed to be:

- Scalable 📈
- Flexible 🔄

- Secure 🔐
- Adapted to local market needs 🌍

🚀 Project Development Plan